

# Closed-Loop Consensus Fusion of Beamformers and NMF Priors for Cocktail-Party Source Separation

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**Abstract**—Mask-informed beamforming couples spectral and spatial cues by letting a time–frequency (TF) mask supply spatial covariance matrices to a beamformer, but the coupling is open-loop: the beamformer output never revises the mask that produced it. We propose DCB-NMF, a closed-loop separator that treats nonnegative matrix factorization (NMF) as a *fusion operator* rather than only a mask estimator. For every source we collect several nonnegative views—the MVDR output, a single-channel NMF prior, the masked MVDR output, and an optional LCMV output nulling the competing talkers—factorize them jointly under a *shared* activation matrix, and read the estimate out as the mean of a designated *consensus subset* of the per-view reconstructions. The fused magnitudes are renormalized into masks and re-enter covariance estimation, closing the loop. On a controlled four-microphone cocktail-party grid (three angular separations  $\times$  three temporal-overlap ratios, 45 test mixtures, with all design choices fixed on a disjoint development set), DCB-NMF reaches  $15.19 \pm 1.89$  dB SI-SDRi versus  $9.23 \pm 8.77$  dB for mask-informed MVDR—a 5.95 dB mean gain (Wilcoxon  $p = 2.1 \times 10^{-7}$ ) and a  $4.6\times$  reduction in variance—and is the best of twelve separators in all nine conditions. An ablation isolates the closed loop as the decisive component ( $-2.89$  dB when removed) and reports a negative result we consider informative: with well-estimated masked covariances, the explicit LCMV nulls are redundant ( $-0.03$  dB when removed), and are harmful if admitted into the readout ( $-3.89$  dB).

**Index Terms**—speech separation, cocktail-party problem, beamforming, MVDR, LCMV, nonnegative matrix factorization, spatial covariance matrix

## I. INTRODUCTION

Separating concurrent talkers recorded by a microphone array—the cocktail-party problem [1]—remains a core task in multichannel front-ends. *Spectral* cues exploit the low-rank harmonic structure of speech and are captured by nonnegative matrix factorization (NMF) [2]–[6] or learned mask estimators [7], [8]; *spatial* cues exploit propagation delays and are captured by beamformers such as MVDR [9], [10], LCMV [11], [12] or the multichannel Wiener filter [13]. Neither suffices alone: NMF masks are fragile when talkers overlap in time and frequency, while a purely spatial filter cannot resolve sources sharing a direction or a formant region [14].

The dominant way to combine them is *mask-informed beamforming*: a spectral mask weights the observations to estimate target and interference-plus-noise spatial covariance matrices (SCMs), from which a single MVDR filter is de-

rived [15]–[17]. Two structural properties of this design motivate the present work.

(i) **The coupling is open-loop.** The mask informs the beamformer, but the beamformer never informs the mask, so any error in the initial clustering of NMF bases into sources propagates unchecked into the SCMs and every downstream stage—worst under high temporal overlap, precisely where the mask is least reliable.

(ii) **Combination is multiplication, not inference.** The standard postfilter multiplies the beamformer output by the mask: a TF bin suppressed by either factor is lost, and two disagreeing sources of evidence have no way to reconcile.

We propose **DCB-NMF**, which addresses both. Rather than multiplying the beamformer output by a mask, we assemble several nonnegative *views* of the same source—MVDR output, spectral prior, masked MVDR, and an optional LCMV output nulling the competing talkers—and factorize them *jointly* under a shared activation matrix, so a temporal pattern must fit the spatial and spectral evidence at once to receive energy. The estimate is read out as the mean over a designated *consensus subset* of the per-view reconstructions, decoupling “which views inform the model” from “which views produce the output”: an informative but distorted view can shape the activations without contaminating the estimate. The fused magnitudes then re-enter SCM estimation, closing the loop. Unlike MNMF-informed beamforming [17], which supplies SCMs to one beamformer and typically enhances one speaker, DCB-NMF runs on every source and uses NMF as fusion operator as well as mask estimator.

Our contributions are: (1) a shared-basis NMF fusion rule with consensus-subset readout; (2) a closed-loop mask update tying the fused output back into SCM estimation; (3) a controlled evaluation over the joint (angular separation  $\times$  temporal overlap) grid, with all design choices fixed on a disjoint development set; and (4) an ablation that attributes the gain honestly—the closed loop dominates, while the explicit LCMV nulls, though formed in every test condition, prove redundant given well-estimated masked covariances.

## II. SIGNAL MODEL AND BACKGROUND

With  $N$  talkers,  $M$  microphones, and short-time Fourier transform (STFT) indices  $f \in \{1, \dots, F\}$ ,  $t \in \{1, \dots, T\}$ , the

observation vector  $\mathbf{x}(f, t) \in \mathbb{C}^M$  is

$$\mathbf{x}(f, t) = \sum_{k=1}^N \mathbf{a}_k(f) s_k(f, t) + \mathbf{n}(f, t), \quad (1)$$

where  $s_k(f, t)$  is the  $k$ -th source at the reference microphone,  $\mathbf{a}_k(f)$  its acoustic transfer function, and  $\mathbf{n}(f, t)$  an additive diffuse noise field. For a far-field source arriving from  $\theta$  on a linear array with microphone positions  $p_m$ , the steering vector is  $[\mathbf{a}(f, \theta)]_m = \exp(-j2\pi f p_m \sin \theta / c)$ , phase-normalized so the reference microphone is real and nonnegative.

Given a TF mask  $\mathcal{M}_k(f, t) \in [0, 1]$ , the masked SCM is

$$\Phi_k(f) = \frac{\sum_t \mathcal{M}_k(f, t) \mathbf{x}(f, t) \mathbf{x}(f, t)^H}{\sum_t \mathcal{M}_k(f, t)}, \quad (2)$$

and the interference-plus-noise SCM  $\Phi_{\bar{k}}(f)$  is obtained the same way from  $\mathcal{M}_{\bar{k}} = \min(\sum_{j \neq k} \mathcal{M}_j, 1)$ . All inversions are preceded by diagonal loading [18],  $\Phi \leftarrow \Phi + \lambda \text{tr}(\Phi) M^{-1} \mathbf{I}$ .

### III. PROPOSED METHOD: DCB-NMF

DCB-NMF proceeds in four stages, the last three of which are iterated (Fig. 1).

#### A. Stage 1: NMF initialization and permutation alignment

We factorize the reference-microphone magnitude spectrogram  $\mathbf{V} = |\mathbf{X}_{\text{ref}}| \in \mathbb{R}_+^{F \times T}$  with KL-divergence NMF [3],  $\mathbf{V} \approx \mathbf{W}\mathbf{H}$ ,  $\mathbf{W} \in \mathbb{R}_+^{F \times K}$ ,  $\mathbf{H} \in \mathbb{R}_+^{K \times T}$ , by multiplicative updates.

The  $K$  bases must be grouped into  $N$  sources. We cluster the columns of  $\mathbf{W}$ ,  $\ell_2$ -normalized, by spherical  $k$ -means, and not the rows of  $\mathbf{H}$ : in a cocktail-party mixture the talkers are simultaneously active by construction, so their activations are strongly correlated, whereas the harmonic and formant structure that distinguishes them lives in the spectral templates. With  $\mathcal{I}_k$  the bases assigned to source  $k$ ,

$$\mathcal{M}_k = \frac{\mathbf{W}_{:, \mathcal{I}_k} \mathbf{H}_{\mathcal{I}_k, :}}{\sum_j \mathbf{W}_{:, \mathcal{I}_j} \mathbf{H}_{\mathcal{I}_j, :}}, \quad \mathbf{P}_k = \mathbf{W}_{:, \mathcal{I}_k} \mathbf{H}_{\mathcal{I}_k, :}. \quad (3)$$

If coarse DOA priors  $\{\theta_k\}$  are available, the mask-to-source assignment is resolved by inter-microphone phase-difference (IPD) agreement. With  $\phi(f, t) = \angle(X_1(f, t)X_0^*(f, t))$  the observed IPD and  $\phi_\theta(f)$  the IPD predicted by the manifold at  $\theta$ ,

$$\mathcal{S}(k, \theta) = \sum_{f, t} \frac{\mathcal{M}_k(f, t)}{\sum_{f', t'} \mathcal{M}_k(f', t')} \cos(\phi(f, t) - \phi_\theta(f)), \quad (4)$$

and masks are greedily assigned to directions by maximizing  $\mathcal{S}$ . Absent a prior, the DOA is estimated from  $\Phi_k$  by steered-response power over a  $1^\circ$  grid.

#### B. Stage 2: parallel spatial filters

For each source  $k$  we form two spatial filters from the same statistics. The steering vector  $\mathbf{a}_k$  comes from the *array manifold* at the estimated or prior DOA rather than the principal eigenvector of  $\Phi_k$ , which is far less sensitive to mask errors, at the price of a far-field assumption.

#### MVDR.

$$\mathbf{w}_k^{\text{MV}}(f) = \frac{\Phi_k^{-1}(f) \mathbf{a}_k(f)}{\mathbf{a}_k^H(f) \Phi_k^{-1}(f) \mathbf{a}_k(f)}. \quad (5)$$

**Geometry-gated LCMV.** Let  $\mathcal{J}_k$  collect the competing sources  $j \neq k$  passing the steering-coherence gate

$$\gamma(k, j) = \frac{1}{F} \sum_f \frac{|\mathbf{a}_k^H(f) \mathbf{a}_j(f)|}{\|\mathbf{a}_k(f)\| \|\mathbf{a}_j(f)\|} < \tau, \quad (6)$$

truncated to at most  $M - 2$  interferers so that  $\mathbf{C}_k = [\mathbf{a}_k, \{\mathbf{a}_j\}_{j \in \mathcal{J}_k}]$  stays well conditioned. With  $\mathbf{g} = [1, 0, \dots, 0]^T$ ,

$$\mathbf{w}_k^{\text{LC}}(f) = \Phi_k^{-1} \mathbf{C}_k (\mathbf{C}_k^H \Phi_k^{-1} \mathbf{C}_k)^{-1} \mathbf{g}. \quad (7)$$

If  $\mathcal{J}_k = \emptyset$ —every interferer too collinear with the target for a stable null—the LCMV branch is disabled and the method falls back to MVDR alone. The outputs are  $Y_k^{\text{MV}} = \mathbf{w}_k^{\text{MVH}} \mathbf{x}$  and  $Y_k^{\text{LC}} = \mathbf{w}_k^{\text{LCH}} \mathbf{x}$ .

#### C. Stage 3: shared-basis fusion with consensus readout

The beamformers, the spectral prior, and their product give four nonnegative views of the same source:

$$\begin{aligned} \mathbf{V}_k^{(1)} &= |Y_k^{\text{MV}}|, & \mathbf{V}_k^{(2)} &= \mathbf{P}_k, \\ \mathbf{V}_k^{(3)} &= \mathcal{M}_k \odot |Y_k^{\text{MV}}|, & \mathbf{V}_k^{(4)} &= |Y_k^{\text{LC}}|. \end{aligned} \quad (8)$$

Rather than combining them by a fixed scalar weight, we stack them vertically,  $\tilde{\mathbf{V}}_k = [\mathbf{V}_k^{(1)}; \dots; \mathbf{V}_k^{(R)}] \in \mathbb{R}_+^{RF \times T}$ , and factorize the stack jointly,

$$\tilde{\mathbf{V}}_k \approx \tilde{\mathbf{W}}_k \tilde{\mathbf{H}}_k, \quad \tilde{\mathbf{W}}_k = [\mathbf{W}_k^{(1)}; \dots; \mathbf{W}_k^{(R)}], \quad (9)$$

so all views share one activation matrix  $\tilde{\mathbf{H}}_k \in \mathbb{R}_+^{B \times T}$  while each keeps its own dictionary  $\mathbf{W}_k^{(r)} \in \mathbb{R}_+^{F \times B}$ . Sharing  $\tilde{\mathbf{H}}_k$  is how the views correct one another: a temporal pattern must fit the spatial *and* the spectral evidence to receive energy.

The per-view reconstructions are  $\hat{\mathbf{V}}_k^{(r)} = \mathbf{W}_k^{(r)} \tilde{\mathbf{H}}_k$ . Crucially, the readout need not use all of them. We average over a *consensus subset*  $\mathcal{C} \subseteq \{1, \dots, R\}$ ,

$$\hat{\mathbf{G}}_k = \begin{cases} \frac{1}{|\mathcal{C}|} \sum_{r \in \mathcal{C}} \hat{\mathbf{V}}_k^{(r)}, & \text{(arithmetic)} \\ \exp\left(\frac{1}{|\mathcal{C}|} \sum_{r \in \mathcal{C}} \log \hat{\mathbf{V}}_k^{(r)}\right), & \text{(geometric)}. \end{cases} \quad (10)$$

Decoupling  $\mathcal{C}$  from the factorized views is the key degree of freedom: a view may shape  $\tilde{\mathbf{H}}_k$  while staying out of the output—relevant for LCMV, whose nulls are informative but whose white-noise gain is not. The geometric readout, an arithmetic mean in the log domain, is the more conservative of the two: a bin survives only if *every* consensus view supports it. Neither choice is obvious a priori, so we fix both on a disjoint development set (Sec. V-A) and report alternatives in Table II.

A final scalar blend with the masked-MVDR view controls how far the estimate may depart from the beamformer,

$$\hat{\mathbf{G}}_k \leftarrow \alpha \hat{\mathbf{G}}_k + (1 - \alpha) \mathbf{V}_k^{(3)}, \quad \alpha \in [0, 1], \quad (11)$$

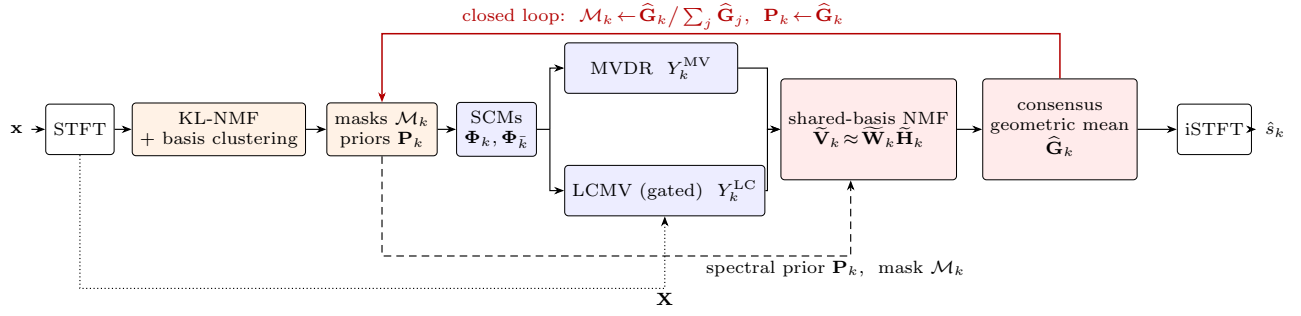


Fig. 1. DCB-NMF. Spectral stages (orange) supply masks and priors; spatial stages (blue) supply beamformer outputs; the fusion stage (red) factorizes all views under a shared activation matrix and reads out the estimate from a consensus subset. The red path is the closed loop, which the ablation identifies as the decisive component.

where  $\alpha = 1$  is pure fusion and  $\alpha = 0$  removes the fusion readout while keeping the closed loop. The complex estimate takes the MVDR phase,  $\hat{S}_k = \hat{\mathbf{G}}_k \odot \exp(j\angle Y_k^{\text{MV}})$ .

#### D. Stage 4: closed-loop mask update

The fused magnitudes are renormalized into masks and become the priors of the next outer iteration,

$$\mathcal{M}_k \leftarrow \frac{\hat{\mathbf{G}}_k}{\sum_j \hat{\mathbf{G}}_j}, \quad \mathbf{P}_k \leftarrow \hat{\mathbf{G}}_k, \quad (12)$$

which re-enters (2). Because the updated mask now reflects spatial as well as spectral evidence, the next iteration's SCMs are cleaner than those from single-channel NMF alone.  $L = 2$  outer iterations are used throughout.

#### E. Computational cost

Per outer iteration and per source, cost is dominated by SCM accumulation,  $\mathcal{O}(FTM^2)$ , the  $F$  linear solves,  $\mathcal{O}(FM^3)$ , and the fusion NMF,  $\mathcal{O}(RFTB)$  per multiplicative update. With  $M = 4$ ,  $R = 4$ ,  $F = 513$ ,  $T \approx 190$ ,  $B = 10$  the method runs at roughly  $0.4 \times$  real time in unoptimized NumPy on one CPU core, the same order as the mask-informed MVDR baseline it replaces, and without any trained model.

### IV. EXPERIMENTAL SETUP

#### A. Data and array

To control the two factors that determine when spatial and spectral cues help—how far apart the talkers are, and how much they overlap in time—we evaluate on a fully controlled synthetic benchmark rather than a fixed corpus. Two speech-like talkers are generated as harmonic sources with distinct fundamentals ( $f_0 = 110$  and  $200$  Hz) and distinct three-formant envelopes, modulated by a syllabic amplitude envelope, then placed symmetrically about broadside on a uniform linear array of  $M = 4$  omnidirectional microphones with 5 cm spacing, mixed in the far field with fractional-sample delays, power-equalized at the reference microphone, and corrupted by an approximately isotropic diffuse noise field at 10 dB SNR. Signals are 3 s at 16 kHz; the STFT uses a 1024-point Hann window with 256-sample hop.

#### B. Condition grid and data splits

We sweep the angular separation  $\Delta\theta \in \{90^\circ, 120^\circ, 150^\circ\}$  and the temporal-overlap ratio  $\rho \in \{0.25, 0.50, 0.75\}$ , the Jaccard overlap  $|A \cap B|/|A \cup B|$  of the two binary activity patterns, realized within 0.01 of target. The *test* grid uses 5 trials per cell (45 mixtures); a disjoint *development* grid of 3 trials per cell (27 mixtures, non-overlapping seeds) is used—and used only—to fix the readout and blend weight. No test mixture influences any design choice.

#### C. Baselines

All baselines share the *same* Stage-1 NMF masks, DOA estimates and STFT settings, so differences reflect only the spatial filtering and fusion strategy: single-channel Wiener NMF; delay-and-sum (**DS**); **Ratio+DS** (NMF mask on the DS output); **MPDR**/Capon on the mixture SCM [9]; **MVDR** (5); **LCMV** (7); **GEV** max-SNR [19]; rank-1 **MWF** [13]; zero-forcing (**ZF**); **AuxIVA** [20] after PCA pre-whitening; and **MVDR+Mask**, MVDR followed by an NMF soft-mask postfilter—the standard mask-informed design [15], [16] and our primary comparison.

#### D. Metric

We report the scale-invariant signal-to-distortion ratio improvement [21],  $\text{SI-SDRi} = \text{SI-SDR}(\hat{s}) - \text{SI-SDR}(x_{\text{ref}})$ , against the dry spatial image of each source at the reference microphone, evaluated permutation-invariantly; the unprocessed mixture scores 0 dB by construction. Significance uses a two-sided Wilcoxon signed-rank test over the 45 paired test mixtures.

### V. RESULTS

#### A. Development-set selection of the readout

We swept both readouts of (10) and the blend weight  $\alpha$  on the development set. The arithmetic readout with  $\mathcal{C} = \{1, 2, 3\}$ —LCMV factorized but excluded from the output—is best, peaking at  $\alpha^* = 0.4$  with 15.77 dB; the geometric readout reaches 15.44 dB with  $\mathcal{C} = \{1, 2, 3\}$  and 15.59 dB with  $\mathcal{C} = \{1, 2, 3, 4\}$ . The three well-behaved variants thus differ by less than 0.4 dB, whereas admitting LCMV into an *arithmetic* consensus is catastrophic (10.68 dB, i.e.  $-5.0$  dB), already

TABLE I

CONDITION-WISE MEAN SI-SDRi (dB) ON THE TEST GRID: 5 TRIALS PER CELL, 45 MIXTURES IN TOTAL, 4-MICROPHONE UNIFORM LINEAR ARRAY, 10 dB DIFFUSE-NOISE SNR. COLUMNS ARE ANGULAR SEPARATION  $\Delta\theta \times$  TEMPORAL OVERLAP  $\rho$ . EVERY METHOD SHARES THE SAME STAGE-1 NMF MASKS, DOA ESTIMATES AND STFT SETTINGS. DCB-NMF USES THE SINGLE CONFIGURATION SELECTED ON A DISJOINT DEVELOPMENT SET (SEC. V-A); NO PER-CONDITION TUNING IS PERFORMED. BEST PER COLUMN IN BOLD; THE LAST COLUMN IS THE MARGINAL MEAN  $\pm$  STANDARD DEVIATION OVER ALL 45 MIXTURES.

| Method                | $\Delta\theta = 90^\circ$ |              |              | $\Delta\theta = 120^\circ$ |              |              | $\Delta\theta = 150^\circ$ |              |              | All                               |
|-----------------------|---------------------------|--------------|--------------|----------------------------|--------------|--------------|----------------------------|--------------|--------------|-----------------------------------|
|                       | $\rho=0.25$               | $\rho=0.50$  | $\rho=0.75$  | $\rho=0.25$                | $\rho=0.50$  | $\rho=0.75$  | $\rho=0.25$                | $\rho=0.50$  | $\rho=0.75$  |                                   |
| NMF (single-channel)  | 12.39                     | 0.27         | -0.87        | 7.93                       | -0.75        | 0.33         | 8.96                       | 4.07         | 1.71         | $3.78 \pm 10.2$                   |
| DS                    | 4.54                      | 4.54         | 4.56         | 5.34                       | 5.32         | 5.31         | 5.75                       | 5.76         | 5.77         | $5.21 \pm 0.5$                    |
| Ratio+DS              | 14.80                     | 1.28         | -0.23        | 10.22                      | 1.17         | 2.77         | 11.17                      | 5.71         | 4.56         | $5.72 \pm 10.9$                   |
| MPDR                  | 10.70                     | 10.58        | 10.68        | 10.80                      | 10.81        | 10.71        | 11.13                      | 11.03        | 11.10        | $10.84 \pm 0.2$                   |
| MVDR                  | 11.46                     | 11.06        | 10.98        | 11.72                      | 11.33        | 11.03        | 12.18                      | 11.98        | 11.79        | $11.50 \pm 0.5$                   |
| LCMV                  | 1.67                      | -5.56        | -7.91        | -4.31                      | 3.43         | 5.35         | -3.22                      | 1.23         | 2.75         | $-0.73 \pm 7.2$                   |
| GEV                   | 1.78                      | -13.32       | -14.73       | -5.54                      | -14.44       | -16.10       | -12.54                     | -17.22       | -16.52       | $-12.07 \pm 11.1$                 |
| MWF                   | 12.16                     | 3.10         | 1.43         | 9.66                       | 2.08         | 2.21         | 9.50                       | 5.77         | 3.08         | $5.44 \pm 7.4$                    |
| ZF                    | 1.67                      | -1.13        | -2.52        | -3.31                      | 1.79         | 3.26         | -2.20                      | 0.73         | 1.88         | $0.02 \pm 4.0$                    |
| AuxIVA                | -8.90                     | -15.57       | -15.80       | -11.60                     | -15.68       | -14.47       | -10.46                     | -13.29       | -14.47       | $-13.36 \pm 5.9$                  |
| MVDR+Mask             | 16.07                     | 5.26         | 4.34         | 12.24                      | 5.75         | 7.55         | 14.16                      | 9.34         | 8.41         | $9.23 \pm 8.8$                    |
| <b>DCB-NMF (ours)</b> | <b>16.26</b>              | <b>14.24</b> | <b>14.06</b> | <b>16.16</b>               | <b>14.34</b> | <b>14.08</b> | <b>16.63</b>               | <b>15.69</b> | <b>15.23</b> | <b><math>15.19 \pm 1.9</math></b> |

foreshadowing the ablation below. This single configuration is used for *every* test result that follows; no per-condition tuning is performed.

### B. Condition-wise comparison

Table I reports mean SI-SDRi in each cell of the (separation  $\times$  overlap) grid. DCB-NMF is the best method in all nine conditions. Over the full test set it reaches  $15.19 \pm 1.89$  dB against  $9.23 \pm 8.77$  dB for MVDR+Mask (a  $+5.95$  dB mean gain, Wilcoxon  $p = 2.1 \times 10^{-7}$ ) and  $11.50 \pm 0.49$  dB for plain MVDR ( $+3.68$  dB,  $p = 5.7 \times 10^{-14}$ , winning on 45/45 mixtures).

The structure of the gain matters more than its magnitude (Fig. 2(a)). At low overlap ( $\rho = 0.25$ ) the masks are reliable and MVDR+Mask is strong (14.15 dB, only 2.20 dB behind). As overlap grows the masks degrade and MVDR+Mask collapses to 6.78 and 6.77 dB while DCB-NMF holds at 14.75 and 14.45 dB—gaps of 7.97 and 7.69 dB. The multiplicative postfilter is what fails: when the mask is wrong, multiplying by it destroys the beamformer output, which is why MVDR+Mask falls *below* plain MVDR at high overlap.

Variance tells the same story. MVDR+Mask has a standard deviation of 8.77 dB and is occasionally worse than the unprocessed mixture (worst case  $-5.99$  dB); DCB-NMF’s is 1.89 dB, a  $4.6\times$  reduction, with a worst case of  $+12.4$  dB. It raises the median only modestly (16.27 vs 14.89 dB) but removes the lower tail entirely. Plain MVDR is stable ( $\pm 0.49$  dB) yet plateaus near 11.5 dB: mask-based methods buy the extra headroom, and DCB-NMF is the only one that buys it without the fragility.

Performance is a broad, single-peaked function of  $\alpha$  (Fig. 2(b)): any value in  $[0.2, 0.6]$  lies within 0.15 dB of the best. The informative endpoint is  $\alpha = 0$ , which disables the fusion readout but keeps the closed loop: it loses 1.62 dB in mean and doubles the standard deviation (3.74 vs 1.89 dB)—the fusion readout contributes less to the average than to the reliability.

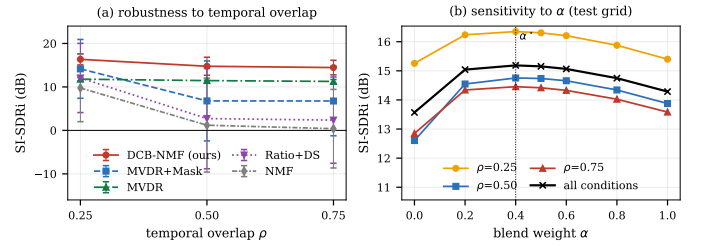


Fig. 2. (a) SI-SDRi versus temporal overlap (mean  $\pm$  std over angular separations and trials). Mask-informed beamforming degrades sharply with overlap; DCB-NMF does not. (b) Sensitivity to the blend weight  $\alpha$  on the test grid;  $\alpha^* = 0.4$  was fixed on the development set. The optimum is broad.

### C. Ablation

Table II removes one component at a time at  $\alpha^*$ . The results are not uniformly flattering to the design, and we report them as measured.

The **closed loop is decisive**: a single outer iteration costs 2.89 dB, concentrated where the initial masks are worst ( $-1.01$  dB at  $\rho = 0.25$  versus  $-4.00$  dB at  $\rho = 0.75$ ). This is the component that distinguishes DCB-NMF from open-loop mask-informed beamforming, and it behaves as Sec. I predicts. The **consensus readout** matters moderately: the geometric variant costs 0.81 dB, its conservatism over-pruning bins where the spectral prior is weak but the spatial evidence is strong. Two components, however, do *not* pay for themselves. Independent per-view factorizations cost 0.06 dB, and—more strikingly—**removing the LCMV branch entirely costs 0.03 dB**, indistinguishable from zero, even though the gate (6) opened and the filter was formed in all 45 test mixtures; moving LCMV into the consensus subset costs 3.89 dB.

### D. Why the LCMV branch does not pay

The LCMV result invites a direct interpretation. An explicit null forces the array response to vanish towards the interferer, but  $\Phi_{\bar{k}}$  in (5) is already estimated from the interferer’s own

TABLE II

ABLATION ON THE TEST GRID AT  $\alpha^*=0.4$ , SPLIT BY TEMPORAL OVERLAP.  $\Delta$  IS THE CHANGE IN THE ALL-CONDITION MEAN RELATIVE TO THE FULL METHOD. ONLY THE CLOSED LOOP IS DECISIVE; THE READOUT DETAILS MATTER BY LESS THAN 1 dB.

| Variant                   | overlap $\rho$ |       |       |       | $\Delta$ |
|---------------------------|----------------|-------|-------|-------|----------|
|                           | 0.25           | 0.50  | 0.75  | All   |          |
| <b>DCB-NMF (full)</b>     | 16.35          | 14.75 | 14.45 | 15.19 | —        |
| w/o closed loop ( $L=1$ ) | 15.34          | 11.10 | 10.45 | 12.30 | -2.89    |
| w/o LCMV branch           | 16.32          | 14.72 | 14.42 | 15.15 | -0.03    |
| w/o shared basis          | 16.30          | 14.70 | 14.39 | 15.13 | -0.06    |
| geometric readout         | 16.10          | 13.46 | 13.57 | 14.38 | -0.81    |
| LCMV inside $\mathcal{C}$ | 11.75          | 11.38 | 10.77 | 11.30 | -3.89    |

mask, so MVDR places a data-driven null in essentially the same place—softer, yet without spending a degree of freedom or incurring the white-noise gain of a hard constraint. Once the masks make  $\Phi_{\bar{k}}$  a faithful interference covariance, the explicit constraint is redundant, and it turns harmful only once its output reaches the readout. The value of the second beamformer is thus not the filter but the *discipline of excluding it*: the consensus subset is what makes an unreliable view safe to keep in the model. LCMV may earn its place in regimes this benchmark omits—more sources, or non-stationary interferers—but we have not shown this.

Three limitations should be stated plainly. The evaluation uses *synthetic*, anechoic far-field mixtures with speech-like harmonic sources: this isolates the factors of interest but does not establish performance on recorded speech, and the far-field manifold assumption behind (5)–(7) fails under reverberation—moving to relative transfer functions [12] is the next step. All experiments use  $N = 2$  sources with a known source count; the formulation supports  $N > 2$  (up to  $M - 1$  nulls) but this is untested. Finally,  $\mathcal{C}$  and  $\alpha$  are fixed globally.

## VI. CONCLUSION

DCB-NMF uses NMF as a fusion operator: nonnegative views of each source—beamformer outputs and spectral priors—are factorized under a shared activation matrix, the estimate is read out from a consensus subset of the reconstructions, and the result is fed back into covariance estimation. With all design choices fixed on a disjoint development set, it is the best of twelve separators in all nine conditions of a controlled cocktail-party grid, improving mean SI-SDRi by 5.95 dB over mask-informed MVDR while reducing its variance 4.6 $\times$ ; the advantage grows with temporal overlap, precisely where multiplicative mask postfiltering fails. The ablation attributes the gain principally to the closed loop and finds the explicit LCMV nulls redundant under well-estimated masked covariances—a negative result delimiting when a second, hard-constrained beamformer is worth its cost. Future work targets reverberant and recorded speech, more talkers, and adaptive consensus selection.

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